Earth and Planetary Science Letters, 27 (1975) 329–341
© Elsevier Scientific Publishing Company, Amsterdam – Printed in The Netherlands

THE PALAEOPOSITION OF MADAGASCAR: PALAEOMAGNETIC EVIDENCE FROM THE ISALO GROUP

BRIAN J.J. EMBLETON* and MICHAEL W. McELHINNY

Research School of Earth Sciences, Australian National University, Canberra, A.C.T. (Australia)

Received April 22, 1975
Revised version received July 7, 1975

Palaeomagnetic results are reported from the continental facies of the Triassic-Jurassic Isalo Group of Madagascar. Stability of the magnetic remanence was tested using the alternating field and progressive thermal demagnetization techniques. Results from 8 sites, 6 located in northwestern Madagascar and 2 from the southwestern region, yield a palaeomagnetic pole at 74.2°S, 97.1°E ($N = 8, A_{95} = 6.3°$). Three models previously proposed to describe the drift history of Madagascar relative to Africa are considered and the relevant geological and geophysical information is reviewed. The palaeomagnetic data are only consistent with the pre-drift model which places Madagascar off the east coast of Africa adjacent to Kenya and Tanzania. This is also the continental drift fit favoured on geological grounds.

1. Introduction

The location of Madagascar in the Gondwana supercontinent during Karroo times has been the subject of considerable discussion for several decades. The exact palaeo-geographical relationship of east and west Gondwanaland partly depends on how Madagascar is reconstructed to the opposing African continental margin. The debate has mainly centered on geological comparisons of east and southeast Africa with Madagascar. There are three possibilities usually considered tenable regarding Madagascar's fit in the reconstruction:

(1) It occupied the Mozambique coastal embayment in southeast Africa – this fit was favoured by Wegener [1] and the geological arguments supporting it have been detailed recently by Flores [2];

(2) Madagascar lay further north off east Africa, adjacent to Kenya and Tanzania – this fit was strongly advocated by Du Toit [3] and was that favoured on geological grounds by Smith and Hallam [4] as the basis for attempting a geometrical reconstruction; or

(3) Madagascar has remained in its present position – a less popular solution but favoured for a time by Tarling [5] who based his preference only on arguments against a northerly fit. Darracott [6] has recently discussed the nature of the crustal structure of southeast Africa and adjacent oceanic regions based on the interpretation of the regional gravity field data – he provides a very useful summary of the current state of knowledge concerning the crustal types between Madagascar and the mainland. His main conclusion is that Madagascar has always occupied its present latitude with respect to Africa, but, in order to explain the anomalous crust (termed thin continental or transitional) between the two, he envisages "some small eastward movement of Madagascar". Results from the Deep Sea Drilling Project have not clarified Madagascar's drift history although the initial report [7] claimed that the data did not support a Cretaceous palaeoposition adjacent to east Africa. Geophysical modelling based on aeromagnetic profiles across the Mozambique Channel led Green [8] to fit Madagascar against Mozambique. His thesis was supported by Kent [9] who believes that with increasing

* Now at: Division of Mineral Physics, CSIRO Minerals Research Laboratories, P.O. Box 136, North Ryde, N.S.W. 2113, Australia.

knowledge of the east African coastal succession, the fit against Kenya seems progressively less probable – although he does admit to the equivocal nature of the stratigraphical evidence. This will be discussed more fully in section 5.

The only previous palaeomagnetic work on the Karroo of Madagascar was carried out by Nairn [10]. Six samples from two sites in the Série Rouge Inférieure, which immediately overlies the coal-bearing strata of the Late Carboniferous–Permian Sakoa Group, yielded consistent directions of magnetization, oblique to the present field direction and reversed (consistent with the polarity of the late Palaeozoic magnetic interval) but based on the stability of natural remanent magnetization not tested by thermal or magnetic cleaning techniques. A new collection of samples from the Sakoa Group and the overyling Late Permian–Triassic Sakamena Group is currently being investigated. Results were also obtained [10] from the Neogene lateritic "cuirasse ferrugineuse" although the beds which comprise the main succession at Col des Tapias, southwest of Ihosy, belong to the Late Triassic–Middle Jurassic Isalo Group – the youngest group constituting the Madagascan Karroo Supergroup. In this paper, palaeomagnetic results from sediments of the Isalo Group also sampled at Col des Tapias and from the continental sequence in the northwest of the island are presented.

2. The Karroo of Madagascar

The geology of Madagascar, recently reviewed by Besairie [11,12], is summarized in Fig. 1. The surface outcrop of the eastern two-thirds of the island consists primarily of Precambrian crystalline basement. The earliest Palaeozoic rocks are probably the Quartzite Series which outcrops in the central areas of the island. On the basis of plant remains and leaf imprints, those beds are tentatively regarded as Devonian to Carboniferous in age.

The Karroo of Madagascar forms a continuous north-south belt of sediment on the western side of the island, eastwards they transgressively overlie the crystalline basement. The facies is mainly continental but a substantial part of the sequence is marine and lagoonal – volcanics are absent. The sequence ranges from late Carboniferous to Middle Jurassic in age and has been divided by Besairie into three groups separated by unconformities:

Isalo Group – uppermost Early Triassic to Middle Jurassic

Sakamena Group – Permo-Triassic

Sakoa Group – Permo-Carboniferous

The Sakoa Group is locally developed in the southwest of the island. The lowest beds constitute the Glacial Series. Tillites are interbedded with black shales, sometimes varved, and subordinate sandstones; the total thickness is about 400 m. Lower Permian Coal Measures follow the glacial beds conformably and consist of sandstones, carbonaceous shales and coal seams often several metres thick. The beds have yielded species of *Glossopteris* and *Gangamopteris*. Occasionally transgressive, the overlying "Série Rouge", formerly the Lower Red Series, consists of an alternating sequence of red shales, light-coloured sandstones and coarse grits up to 1400 m thick. Silicified wood and remains of a primitive reptile have been recovered. The Sakoa Group is terminated with the Vohitolia Limestones, a 10–30 m marine horizon which represents the first marine transgression in the Madagascan Karroo. The fauna includes the brachiopods *Productus* and *Spiriferina* and reef-buiding stromatolites.

In the southwest of Madagascar the Sakoa is followed unconformably and transgressively by the Sakamena Group. It is principally continental but includes marine intercalations. In the extreme north of the island near Diego Suarez, the Sakamena has an entirely marine facies with abundant Permo-Triassic brachiopods, fish and ammonites. In the south and west, Besairie has divided the group into three parts. The Lower Sakamena (Late Permian in age) has a basal conglomerate followed by marine limestones and plant-bearing shales. Remains of *Thinnfeldia* and *Glossopteris* have been recovered but *Gangamopteris* is absent. The overlying shales and sandstones have yielded an amphibian *Rhinesuchus* and the reptiles *Tangasaurus* and *Hovasaurus*. The Lower Sakamena here is about 2500 m thick. The Middle Sakemena, up to 400 m thick, consists of shales, septarian mudstones and limestones with cone-in-cone structures. The continental facies was periodically invaded by the sea, producing lagoonal and brackish conditions. Fish remains enclosed in shaley nodules have been correlated with species found

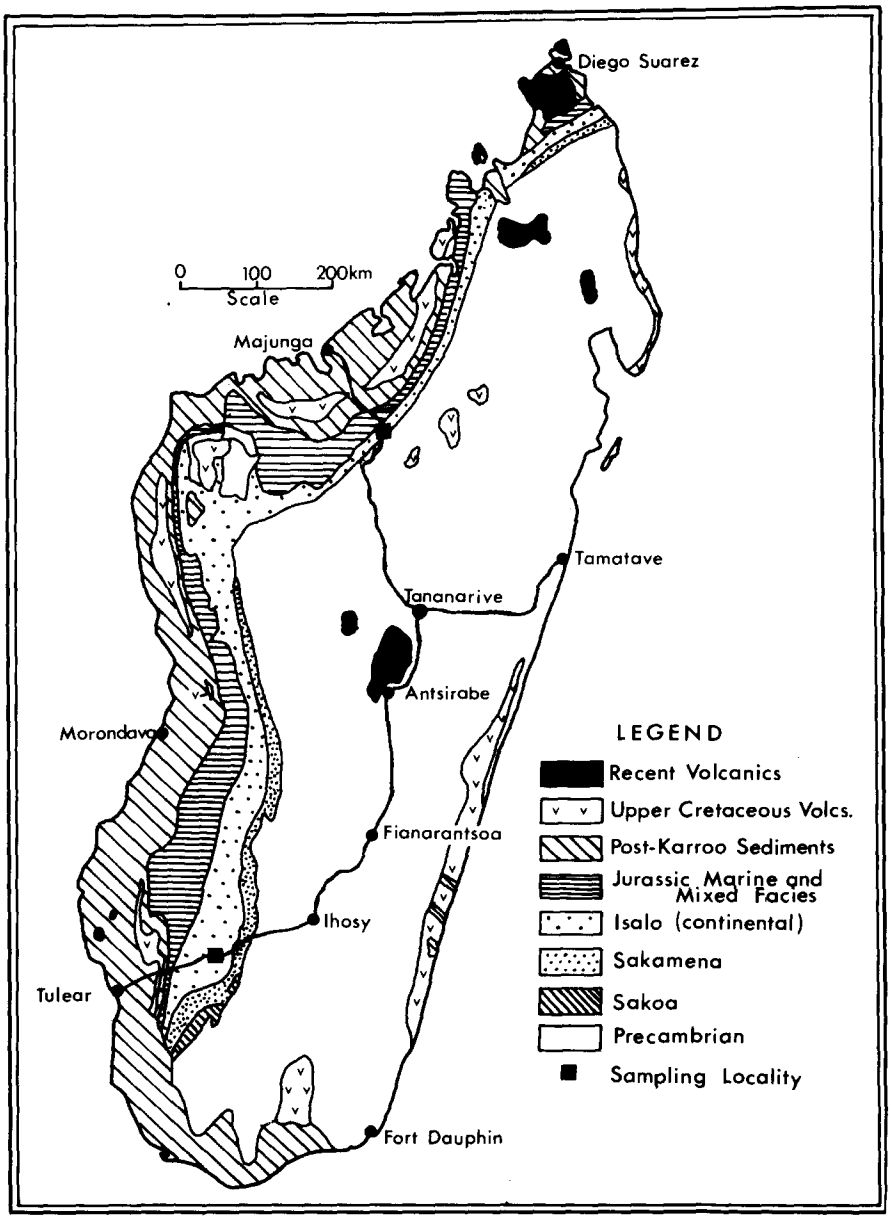


Fig. 1. A simplified geological map of Madagascar modified from Besairie's more detailed version [12]. The Karroo sediments form a narrow belt on the western margin of the island and transgressively overly the Precambrian basement which is undifferentiated in the diagram. Squares indicate the sampling localities in the continental sequence of the Isalo Group.

in the marine sequences further north and indicate an Early Triassic age. Those beds are considered [13] the stratigraphic equivalents to the marine incursion in the Maji ya Chumvi beds of Kenya (see section 5). The upper Sakamena is entirely continental and consists of about 600 m of cross-bedded sandstones with intercalated red shales – formerly the Upper Red Series.

The marine facies of the Sakamena Group in northern Madagascar reaches a thickness of 600 m. The basal sandstones with *Products* and *Spiriferina* lie directly on crystalline basement and are considered by Besairie stratigraphically equivalent to the Vohitolia

Limestones in the southwest. The overlying shaley sandstones contain Late Permian ammonites, particularly *Cyclobus*, preserved in sub-spherical nodules of marly limestone. The lower part of this sequence is correlated with the Lower Sakamena in the southwest. The succeeding beds, mainly shales, contain an extremely rich Early Triassic fish and ammonite fauna. The fossils are enclosed in flattened nodules of calcareous marl. The highest beds of the marine Sakamena Group in northern Madagascar are uppermost Early Triassic in age. They consist of shales and limestones characterized by the presence of the ammonite *Flemingites*, and are correlated with the Upper Sakamena of the southwest.

The Isalo Group unconformably and transgressively overlies the Sakamena Group. The basal members have a wholly continental facies, but marine incursions of long duration occur in the higher beds. The Isalo Group ranges in age from the top of the Early Triassic to the Middle Jurassic (Bathonian); three divisions have been recognized [11]:

Isalo III – shales and sandstones, laterally grade into marine formations;

Isalo II – shales and sandstones with marine intercalations;

Isalo I – continental sandstones.

The sandstones of Isalo I are coarse-grained, often conglomeratic and spectacularly cross-bedded. They were derived directly from the crystalline basement and deposited in fresh water. The beds are almost wholly unfossiliferous except for the flora *Laccopteris* in the highest levels and pelecypod remains in a marine horizon in the north. The thicker succession in the southwest measures about 3000 m.

Isalo II is an alternating sequence of massive buff-coloured sandstones with red shales that are often mottled. In the northwest near Majunga the shales occur in discrete bands which vary from a few centimetres, but rarely exceed half a metre in thickness. In the southern and western parts of the island, the total thickness is about 1000 m. The fossiliferous marine intercalations give an Upper Liassic age. The sandstones and shales of Isalo III represent a facies similar to Isalo II, though marine intercalations are progressively more common. The continental beds contain remains of the dinosaur *Bothriospondylus*.

TABLE 1

Sampling details for the Isalo Group

Site	Samples	Locality	Formation	Rock type
Northwest sites				
1	IS110-113	41.1*	III	red shales
2	IS114-117	24.8*	III	calcareous pink siltstone
3	IS118-120	6.5*	I	mudstone
4	IS121-124	3**	I	mudstone
5	IS125-128	1	I	mudstone
6	IS130A, B, 131	9.0*	I	micaceous chocolate siltstone
7	IS132-135	24.0*	III	red and pink siltstone
Southwest sties				
8	IS22-31	Col des Tapias	II	coarse red sandstones
9	IS102-108	Col des Tapias	II	coarse red sandstones

* Indicates kilometres north of Ambalabongo.
** Indicates successively higher sites (in metres) in mudstone bands sampled 6.5 km north of Ambalabongo.
Site 8 samples from Col des Tapias are from stratigraphically higher beds than site 9 samples – total thickness sampled was about 18 m.

Stagnant lagoonal conditions prevailed during the Bathonian, these highest beds have yielded the pelecypod *Corbula*. Contemporaneous sedimentation in the extreme north of the island proceeded with a dominantly marine facies where limestones and calcareous marls are more abundant.

The Isalo Group was sampled at nine sites in the two localities marked on Fig. 1. Sites 1–7 were situated in outcrops of Isalo I and III north of Ambalabongo along the road to Majunga, in the northwest part of the island. The samples were collected from the shales, mudstones and silts, commonly dark red to chocolate in colour, which usually occur in discrete bands within a massive white to buff-coloured continental sandstone. Dips are generally to the northwest at between 1° and 7°. In the southwest part of the island, sites 8 and 9 were situated on the north and east sides respectively of the Col des Tapias, an escarpment of Isalo II with red horizons in mainly coarse sandstones. That locality is at km 758 on the road from Ranohira to Sakaraha. The escarpment is

approximately 0.5 km south of the road. The regional bedding tilt is 4° to the southwest. Further sampling details are given in Table 1.

3. Palaeomagnetic results

Directions and intensities of remanent magnetism were measured using a DIGICO Balanced Fluxgate Rock Magnetometer [14]. Alternating field (AF) demagnetization was carried out with equipment similar to that described by McElhinny [15] and thermal demagnetization by the progressive method was carried out in a non-magnetic furnace which holds up to 50–25 mm X 25 mm cylindrical specimens. The magnetic field-free environment was maintained with a 10-square coil set as described by McElhinny et al. [16]. Directions of magnetization were analysed using the method devised by Fisher [17] and the two-tier analysis of directions followed the method described by Watson and Irving [18] and McElhinny [19]. The magnetic properties of the red mudstones, siltstones and shales collected in the northwest of the island were different from those properties in the coarser red sediments sampled at Col des Tapias in the southwest; it is appropriate to describe the results separately.

3.1. Northwest sites

Partial thermal demagnetization was carried out on all specimens – normally 4 per sample – at 100°C temperature increments between room temperature and 400°C and at increments of 50°C between 400°C and 650°C. Using the criterion of minimum dispersion (i.e. highest k value for a Fisherian distribution), sample-mean directions of cleaned remanent magnetization were obtained. Those were combined to provide the site-mean directions listed in Table 2 (sites 1–7 inclusive). Similar directions, all normal, were obtained from six of the sites but the samples constituting site 2 yielded anomalous results. At no stage during the partial demagnetization procedure did those directions attempt to align themselves with the main population. The anomaly is not due to field orientation errors since results from four samples are involved, a local magnetic anomaly is also unlikely since samples from site 7 were collected only 0.8 km away, nor was the anomalous direction of magnetization acquired under laboratory conditions since specimens from those particular samples were heated in the furnace with specimens from other samples. For the purposes of the further geophysical calculations based on the "cleaned" data, results from site 2 have been omitted.

TABLE 2

Madagascar data summary – Isalo Group

Site No. (subdivision)	Location	N	D, I	R	α_{95}	Mean VGP		Treatment
						lat. °N	long. °E	
1 (Is. III)	16.4°S, 47.2°E	4	352, −46	3.9615	10	76	259	100–300°C
2 (Is. III)	16.5°S, 47.1°E	4	(72, −24)	3.8805	19	–	–	300°C
3 (Is. I)	16.7°S, 47.1°E	3	344, −40	2.9933	7	74	293	300–400°C
4 (Is. I)	16.7°S, 47.1°E	4	348, −43	3.9882	6	76	278	200–400°C
5 (Is. I)	16.7°S, 47.1°E	4	350, −47	3.9760	8	75	264	100–200°C
6 (Is. I)	16.7°S, 47.1°E	3	340, −37	2.9834	11	71	302	200–300°C
7 (Is. III)	16.5°S, 47.1°E	4	350, −41	3.9758	8	78	277	100–300°C
8 (Is. II)	22.7°S, 45.0°E	5	327, −54	4.7210	21	57	282	600°C, 50 mT/400°C
9 (Is. II)	22.7°S, 45.0°E	5	0, −50	4.8528	15	81	224	600°C, 50 mT/400°C

Mean of site VGPs $N = 8, R = 7.9100, k = 77.7, A_{95} = 6.3$, lat. 74.2°N, long. 277.1°E

Mean of sample VGPs $N = 32, R = 30.9149, k = 28.5, A_{95} = 4.8$, lat. 74.2°N, long. 275.2°E

Sample-mean and site-mean directions of magnetization were subjected to a two-tier analysis to determine the significance of the between-site variation – judged by an F-test.

$$\frac{\text{Mean square between sites}}{\text{Mean square within sites}} = F_{2(B-1),\,2\Sigma(W_i-1)}$$

where "Mean square between sites" = $(\Sigma R_i - R)/2(B-1)$, "Mean square within sites" = $\Sigma(W_i - R_i)/2\Sigma(W_i - 1)$, W_i = number of samples from the ith of B sites, R_i = vector resultant at the ith site, and R = vector sum of all samples.

The analysis, carried out on sites 1 and 3–7, gives $F_{10,32} = 1.767$, where the upper 5% point ($F_{0.05}$) for $F_{10,32} \doteqdot 2.14$. The between-site variation is therefore not significant, i.e. $\beta \approx \infty$ (β = the between-site precision parameter), so that sample means are providing the same time-average as site means. This result suggests two possibilities regarding the time of acquisition of the magnetic remanence other than that the remanence is of primary origin:

(1) that each sample (and the formation) acquired its magnetization over a very short period of time – say during the past 10^2 years; or (2) each sample averages secular variation and was magnetized over a longer period of time – either during the past 10^4 to 10^5 years, or else over a similar period during the Jurassic.

Site directions of magnetization are plotted in Fig. 2B with the mean of sites based on $N = 6$ and its circle of confidence (α_{95}) for $P = 0.05$. Clearly, the mean of the site directions is significantly different from both the present geomagnetic field direction (declination = 350°, inclination = –50°) and the direction of the field due to an axial geocentric dipole (0, –30.8). Therefore explanations supposing that the magnetization is of Recent origin are untenable. The magnetization was probably acquired at the time of deposition or shortly afterwards. That each sample therefore may be regarded as a time average of secular variation can be linked to the effects of slow deposition rates that accompany mudstone and marl formation.

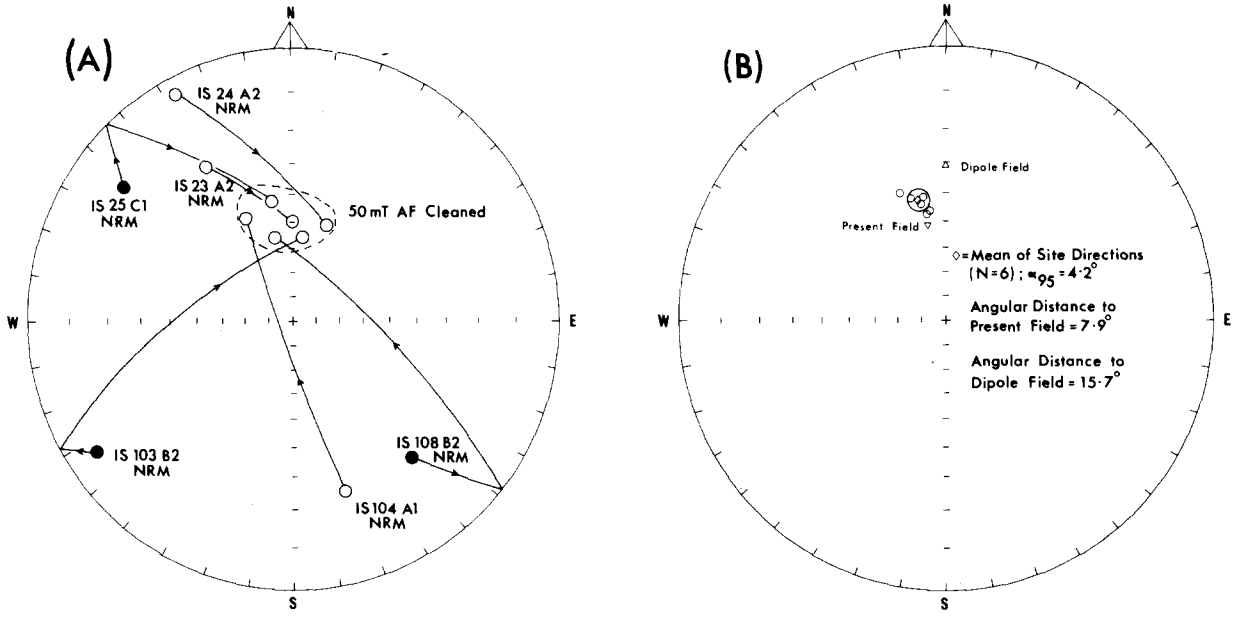


Fig. 2. Scattered directions of natural remanent magnetization are sho vn for specimens from the two southwestern sites (A). Response to treatment in alternating magnetic fields indicated the removal of a randomly oriented component of isothermal remanent magnetization, i.e. such as that acquired due to lightning strike. Fig. 2B shows the northwestern site-mean directions obtained after thermal treatment. The difference between their mean direction and the direction of the present field and dipole field is significant at the 95% probability level. Open symbols represent up-pointing (negative) inclinations and closed symbols represent down-pointing (positive) inclinations. Directions are plotted on a Wulff equal-angle stereographic projection.

3.2. Southwest sites

Directions of natural remanent magnetization at both sites were initially scattered, nor did they appear to plot with a planar distribution towards any preferred axis. Some of the directions obtained before cleaning are shown in Fig. 2A. Initially, two specimens were chosen from each sample for pilot thermal demagnetization using a temperature increment scheme similar to that described above for the northwest sites. Only after treatment at 600°C did the specimen directions approximately align with the mean direction from the northwest sites although during partial thermal demagnetization the mean pilot directions progressively rotated towards that axis. Samples from the two southwest sites were collected from an escarpment outcrop, one of the few topographically prominent features in that district and therefore highly susceptible to lightning strikes. It was decided to partially demagnetize the remainder of the collection in progressively increasing alternating fields although none of the sample-mean intensities seemed excessively high – even for sediments. Site 8 sample intensities ranged from 3.0 to 38.8 mA m^{-1} (sample 27 may have been anomalous at 122 mA m^{-1}) and site 9 sample intensities ranged from 8.1 to 42.8 mA m^{-1}. The spectacular result shown in Fig. 2A was achieved. Relatively low alternating fields – in some cases 5 mT – appeared sufficient to remove a significant proportion of the total initial remanence and directions rotated systematically into the field direction obtained from the sites in the northwest. Little work has been done to compare thermal and AF demagnetization characteristics of rock samples. However, McElhinny and Opdyke [20] have noted that when a large component of isothermal remanent magnetization is present (e.g. that induced by lightning), it is difficult to remove by thermal demagnetization. The results obtained here support their conclusions. The intensities of two specimens from sample 27 were reduced to 3% and 4% of the initial values after treatment in a 50-mT alternating field. A similar intensity level was obtained by thermal demagnetization but only after treatment between 500°C and 600°C. Thus, blocking temperatures for magnetic domains isothermally remagnetised may be sufficiently high to mask the primary magnetic remanence or at best leave a narrow temperature spectrum within which the primary component might be detected. It is also interesting to note the high blocking temperatures for IRM were first described from igneous rocks from which lightning induced IRM may be removed with AF treatment. In the present investigation, although the IRM component was acquired by redbeds, it was likewise relatively "soft" and easily removed with AF treatment.

Pilot AF demagnetization on some specimens showed that the IRM component was removed by 50 mT. All the remaining specimens were, therefore, first demagnetized in a field of 50 mT and then thermally demagnetized to remove secondary chemical remanent magnetizations which may have been acquired subsequent to the primary component. The highest precision was obtained after treatment at 400°C. The results from these specimens were combined with the 600°C results from the specimens used for pilot thermal demagnetization.

Not all samples produced well-grouped specimen directions using this optimum combination of treatments. Specimen directions from 7 of the 17 samples could still be regarded as random [21,22] – these were rejected. The resulting site-mean directions are listed in Table 2 (sites 8 and 9).

All site and sample virtual geomagnetic poles, with their respective means and circles of confidence for $P = 0.05$, are shown in Fig. 3.

The stable magnetic remanence isolated from samples of the Isalo Group, sampled in the northwest and southwest of the island has normal polarity. A worldwide analysis of Mesozoic palaeomagnetic data has previously demonstrated [23] the existence of a predominantly normal period extending from the Late Triassic (about 200 m.y.) to the Late Jurassic (about 150 m.y.).

4. Comparison with Africa

Proposed models concerning Madagascar's drift history have been outlined in section 1. The results described in this paper provide the first reliable estimate of a palaeomagnetic pole position for Madagascar covering a period during which it constituted a part of Gondwanaland. By comparing the data with the data relating to mainland Africa and which cover a similar period of time, it is possible to make a quantitative comparison of the drift models (1) that Madagascar

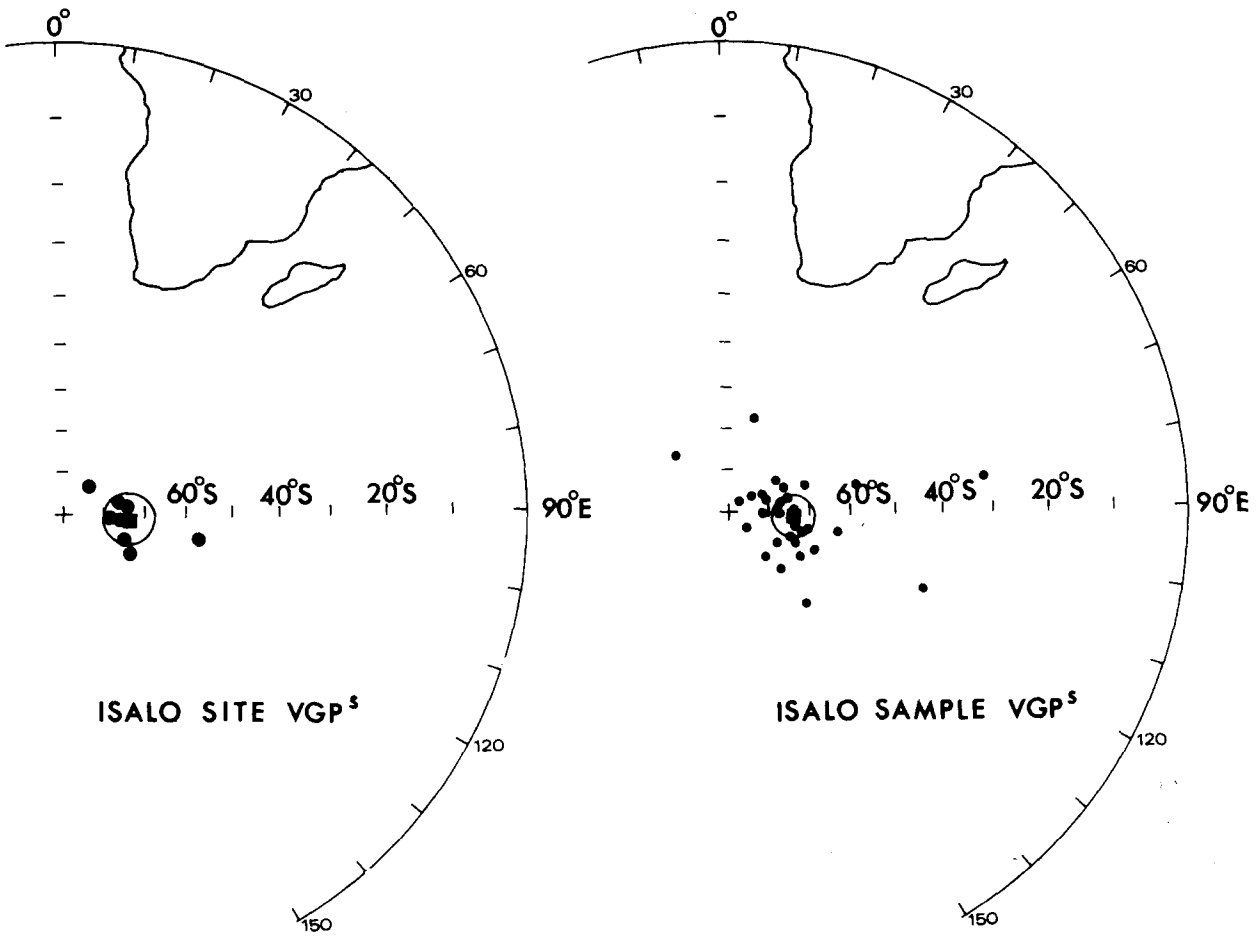


Fig. 3. Site and sample virtual geomagnetic (south) poles obtained from the Isalo Group of Madagascar – plotted on a Wulff equal-angle stereographic projection. Africa and Madagascar are shown in their present relative positions.

TABLE 3

Comparison with various hypotheses

(a) *Least favourable comparison*

African pole for Tru-Jl at 67.4°N, 253.2°E, A_{95} = 5.9°. Madagascar mean site VGPs, A_{95} = 6.3°. Combined errors = 12.2°, significance point at 8.6°

Madagascar pole	Angular difference to African pole	Significant?
Present position (74.2°N, 277.1°E)	10.3°	yes
Smith-Hallam reconstruction (62.5°N, 249.9°E)*	5.0°	no
Flores reconstruction (68.9°N, 290.2°E)**	13.7°	yes

(b) *Most favourable comparison*

African pole for Mesozoic at 67.4°N, 249.5°E, A_{95} = 4.4°. Madagascar mean sample VGPs, A_{95} = 4.8°. Combined errors = 9.2°, significance point at 6.5°

Madagascar pole	Angular difference to African pole	Significant?
Present position (74.2°N, 275.2°E)	10.7°	yes
Smith-Hallam reconstruction (62.4°N, 248.9°E)*	5.2°	no
Flores reconstruction (69.1°N, 288.9°E)**	14.4°	yes

* Pole of rotation 9°S, 313°E by 15°. ** Pole of rotation 48°S, 70°E by 14° (cf. 57.7°S, 157.5°E by 11°, [26]).

occupied the Mozambique coastal embayment in south-east Africa, (2) that it lay further north off east Africa, or (3) that it has remained in its present position. For some time it has been recognized that the latitude of Africa has remained fixed during the whole of the Mesozoic (e.g. [24]). McElhinny and Brock [25] have recently reviewed the African Mesozoic palaeomagnetic data and provide a useful up-to-date summary of

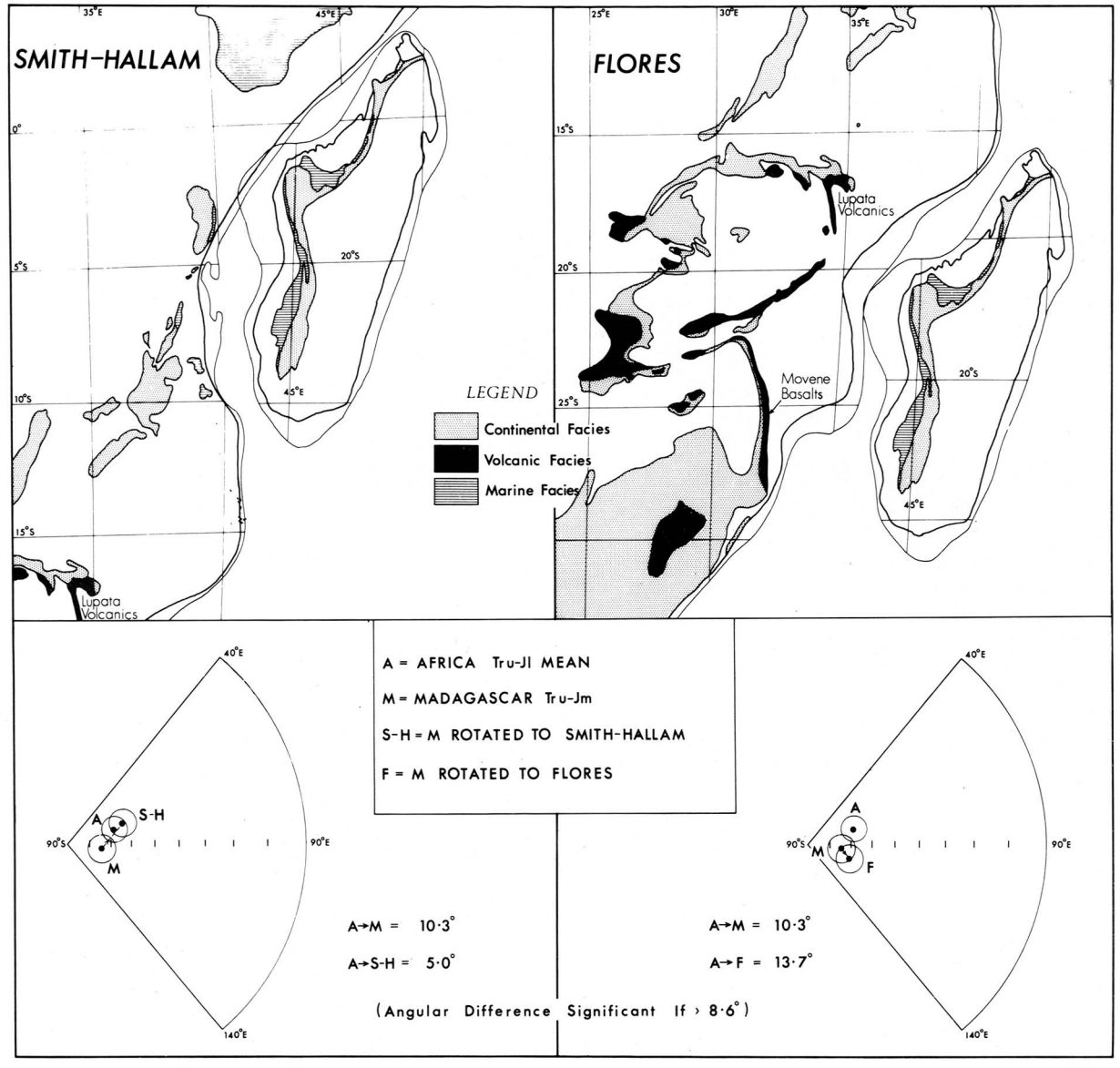


Fig. 4. A comparison of the pre-drift models for Madagascar and Africa. The Smith-Hallam reconstruction is based on a geometrical fit with additional geological constraints [4] and the Flores reconstruction is based on a comparison of the geology of Madagascar with mainland Africa [2]. A simplified geological comparison is attempted in the figure for two models on the basis of the distribution of facies types. The main source for the geological information is the Geological Map of Africa (UNESCO Paris, 1963) modified (1) to distinguish the younger Lupata Volcanics and Movene Basalts from the Karroo Basalts, as done by Flores [2], and (2) to indicate correctly the extent of the igneous province in the Nuanetsi district, Rhodesia. The palaeomagnetic data shown in the lower part of the diagram are based on the "least" favourable comparison (see text and Table 3).

the pole positions. Thus, there are two approaches by which the comparison of data can be made. Firstly, we can compare the Madagascan data with African data that cover as nearly as possible the same time span or secondly, compare them with the African data for the whole of Mesozoic time. Those two methods for presenting the African results and the two methods for presenting the Madagascan results viz. by the mean of site or sample VGPs, suggest two extreme comparisons that can be made, i.e. the "least" and "most" favourable. The results are given in Table 3. Regardless of how the comparison is made, the difference between the palaeomagnetic poles of Africa and Madagascar is not significant *only* for the pre-drift model based on the Smith and Hallam reconstruction [4]. The "least" favourable comparison is shown diagrammatically in Fig. 4.

5. Comments on geological comparisons

Although the stratigraphical evidence related to the drift history of Madagascar has been described as equivocal [9], it is maintained that the Phanerozoic geological history of Madagascar is more similar to that of east Africa than to any other part of the African mainland. Lower Palaeozoic sediments comparable to the Table Mountain Group of south and southeast Africa are not found in Madagascar, where Palaeozoic sedimentation begins with the Upper Carboniferous Sakoa tillite [11]. Kent [27] regards the Karroo sequences in Madagascar as similar to successions found in several of the Karroo basins. It is, therefore, not surprising that strong cases have been presented on geological grounds for fitting Madagascar to Africa and that none of the solutions is considered unique. Du Toit [3], King [28], Dixey [29] and Smith and Hallam [4] have compared the Madagascan Karroo with the Kenyan and Tanzanian sequences, whereas an equally strong case was made by Wellington [30] and later Flores [2] for a comparison with southeast Africa.

Kent [27, p. 874] states that, "the case for a fit of the Karroo of Madagascar with that of any part of the mainland was, however, greatly weakened, if not eliminated, by the documentation of transition of the continental Karroo of Madagascar into a marine facies at several levels", and "this is a feature without parallel in east Africa". However, Miller [13] described the geological succesion of the Mariakani–Mackinnon Road area in Kenya and this section through the Karroo sequence north of Tanga *contains a marine horizon* – it lies between the Late Permian–Early Triassic Lower and Upper Maji ya Chumvi beds. Those beds have been correlated with the Sakamena Group of Madagascar which also contains marine horizons. The marine horizon in east Africa contains nodules enclosing fish scales which have been compared with the Eotriassic marine species *Boreosomus gillioti* (Priem) from Madagascar [13]. Henri Besairie regards the contemporaneous marine incursions as Late Permian (private communication, 1973). Even the underlying plant-bearing horizons of the Lower Maji ya Chumvi beds have yielded specimens similar to some of those from the Sakamena Group and the plant-bearing beds and the fish-bearing beds have the same relative stratigraphical position in both east Africa and Madagascar. The likenesses so impressed Dixey [29, p. 259] that he wrote of "a striking similarity between the Karroo of Madagascar and, for example, that of the Tanganyika coast, as exposed north of Tanga". Further litholigcal comparisons between the Upper Karroo (Jurassic) of east Africa and Madagascar are again evident from the marine facies which succeed early Jurassic continental facies *in both regions.* The presence of an equivalent marine facies in southeast Africa has not been established, although Late Jurassic marine beds have recently been described [31] from south Africa.

Madagascar's affinity with east Africa is also strongly supported on faunal grounds. Smith and Hallam [4] have summarized this evidence for the Jurassic, pointing to the faunal similarities between east Africa Madagascar and Cutch in northwest India, which support the existence of a distinctive faunal sub-province within the Tethyan realm. Haughton [32] makes specific mention of two pelecypods, *Gervillia elianae* and *Modiolopsis stockleyi,* from the lower marine beds of the Sakamena Group. They were first described from the Ruhembe beds of Tanzania. The continental facies of the Sakamena Group have also been correlated with mainland sequences [13] on the basis of the local occurrences of the reptile *Tangasaurus meneli* Haughton.

Flores [2] attempted a comparison of the Karroo lithologies between Madagascar and southeast Africa but was unable to establish faunal comparison between those two regions. The geological argument for the

TABLE 4

Outline geological comparison between Madagascar and east Africa

Period		Madagascar	Unique common features	East Africa
Middle	ISALO	III	Marine facies including esp. Ammonites – *Obtusiositites, Bouleiceras.*	Extensive marine facies
Lower Jurassic	ISALO	II	Continental facies with silicified wood	Mazeras sandstones, Kenya
Upper Triassic	ISALO	I		Mariakani sandstones, Kenya
Lower Triassic	SAKAMENA	6. Sandstones, shales	*Estheria*	Upper Maji ya Chumvi, Kenya
	SAKAMENA	5. Fish and Ammonite bed	*Boreosomus gillioti* (Priem)	Maji ya Chumvi fish bed
Upper Permian	SAKAMENA	4. Shales, sandstones with reptiles	*Tangasaurus meneli*	Tanga beds, Tanzania
	SAKAMENA	3. Plant-bearing shales	*Glossopteris, Voltzia sp*	Lower Maji ya Chumvi beds, Kenya
	SAKAMENA	2. Lower marine bed with pelecypods	*Gervillia elianae, Modiolopsis stockleyi*	Ruhembe beds, Tanzania
	SAKAMENA	1. Basal conglomerate	Conglomerates	Upper Taru Grits, Kenya inc. conglomerates

comparison of Madagascar with east Africa is, in that sense, wholly consistent. A summary of the geological features unique to Madagascar and east Africa is given in Table 4.

The geological comparison of Madagascar with the African mainland would not be complete without considering their respective volcanic histories. In south and southeast Africa, Karroo sedimentation almost ceased with the onset of the Stormberg Volcanic episode. The northern limits of the volcanic outcrop are shown in Fig. 4. No volcanics of this age (Liassic) are known from Madagascar or east Africa. The Jurassic Ferrar dolerites of Antarctica are regarded their nearest equivalents by Smith and Hallam [4] – on their geometrical reconstruction, the two regions are juxtaposed. Extensive volcanic activity began in Madagascar in the Turonian [11] and extended into the Campanian (95–75 m.y.). However, the Lupata and Movene Basalts of Mozambique range from 105 to 135 m.y. [2]. Kent [9] regards these events as nearly contemporaneous and cites them as circumstantial evidence supporting a direct palaeo-geographical relationship. However, the fact that they are not contemporaneous may be equally indicative that they occupied distinct volcanic provinces. It is also likely that the initial rift between Madagascar and the African mainland had taken place by the Cretaceous [2,5,7,33].

Much of the geophysical evidence related to the early evolution of the oceanic regions circumscribing Madagascar is apparently conflicting and ideas require constant revision with the addition of more data. Tracings from seismic reflection records show the existence of a north–south ridge-like feature in the narrowest part of the Mozambique Channel [34]. It may be the remnant of a strike-slip fault active during the early formation of the Indian Ocean. Dextral movement along that line would place Madagascar opposite Somalia rather than against Mozambique to the west. However, on Darracott's model [6], Madagascar has remained in its present latitude relative to Africa, but some eastward movement is still required. He also presents evidence which supports the idea [35] that

the Mozambique Ridge is a fragment of continental material as opposed to an extinct spreading centre. The linear magnetic anomalies in the Mozambique Channel are not oriented north–south [6,36] and do not produce a symmetrical pattern over the ridge crest [37]. Should the linear magnetic anomalies ultimately be referred to an extinct spreading centre, they may equally record the sea-floor spreading episode that initiated Antarctica's split from southeast Africa – a possibility apparently not considered by Green [8]. An originally more northerly position for Madagascar is consistent with the pattern of the linear magnetic anomalies generated during an episode of sea-floor spreading about the Carlsberg Ridge [34]. However, its present position "south" of a pre-drift position against east Africa is considered by Burke and Whiteman [38] to be inconsistent with the sense of more recent (post-anomaly 31) transform motion in the southeast Indian Ocean. The latest motion of Madagascar relative to Africa would seem to have been northward [39]. How that later sense of motion relates to the early drift history of Madagascar remains uncertain. Some of the schemes suggested [40,41] indicate that its drift history may be sufficiently complex such that the evidence from the ocean floor for any early motions may well have been obliterated.

The alternative hypotheses of Madagascar's affinity with the African mainland are summarized in Fig. 4 – a distinction has been made between the facies types within the Karroo Supergroup. The reconstruction of Smith and Hallam [4], following Du Toit [3], is now supported by palaeomagnetic results from Triassic-Jurassic formations from the island and mainland.

Acknowledgements

Our palaeomagnetic study of the Karroo Supergroup of Madagascar was carried out with the cooperation of the Geological Survey, Department of Mines of the Malagasy Republic. We wish to express our appreciation of the willing assistance provided by Le Chef du Service Géologique, Monsieur J. Rasoamahenina and members of his staff. We are indebted to Dr. Henri Besairie for valuable geological discussions and his help in choosing appropriate sampling localities. The assistance of Dr. H.J. Rakotoarivelo and His Excellency T.L. Crosthwait, British Ambassador to the Malagasy Republic, is gratefully acknowledged.

References

1 A. Wegener, The Origin of Continents and Oceans (Methuen and Co., London, 1929) 4th German ed., 248 pp., translation.
2 G. Flores, Suggested origin of the Mozambique Channel, Trans. Geol. Soc. S. Africa 73 (1970) 1–16.
3 A.L. du Toit, Our Wandering Continents (Oliver and Boyd, Edinburgh, 1937) 366 pp.
4 A.G. Smith and A. Hallam, The fit of the southern continents, Nature 225 (1970) 139–144.
5 D.H. Tarling, Gondwanaland, palaeomagnetism and continental drift, Nature 229 (1971) 17–21.
6 B.W. Darracott, On the crustal structure and evolution of southeastern Africa and the adjacent Indian Ocean, Earth Planet. Sci. Lett. 24 (1974) 282–290.
7 The Scientific Staff, Preliminary report of Leg 25 of the Deep Sea Drilling Project – Western Indian Ocean, Geotimes 17 (1972) 21–24.
8 A.G. Green, Sea-floor spreading in the Mozambique Channel, Nature Phys. Sci. 236 (1972) 19–21.
9 P.E. Kent, Mesozoic history of the east coast of Africa, Nature 238 (1972) 147–148.
10 A.E.M. Nairn, Palaeomagnetic measurements on Karroo and post-Karroo rocks. A second progress report, Overseas Geol. Min. Resour. 9 (1964) 302–320.
11 H. Besairie, Geologie de Madagascar, I. Les terrains sédimentaires. Ann. Géol. Madagascar XXXV (1972) 463 pp.
12 H. Besairie, Précis Géologie Malgache, Ann. Géol. Madagascar XXXVI (1973) 143 pp.
13 J.M. Miller, Geology of the Mariakani–Mackinnon Road area, Geol. Surv. Kenya Rep. 20 (1952) 32 pp.
14 L. Molyneux, A complete result magnetometer for measuring the remanent magnetization of rocks, Geophys. J. 24 (1971) 429–433.
15 M.W. McElhinny, An improved method for demagnetizing rocks in alternating magnetic fields, Geophys. J. 10 (1966) 369–374.
16 M.W. McElhinny, G.R. Luck and D. Edwards, A large volume magnetic field free space for thermal demagnetization and other experiments in palaeomagnetism, Pure Appl. Geophys. 90 (1971) 126–130.
17 R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. London, Ser. A 217 (1953) 295–305.
18 G.S. Watson and E. Irving, Statistical methods in rock magnetism, Mon. Not. R. Astron. Soc., Geophys. Suppl. 7 (1957) 289–300.
19 M.W. McElhinny, Statistics of a spherical distribution; in: Methods in Palaeomagnetism, eds. D.W. Collinson, K.M. Creer and S.K. Runcorn (Elsevier, Amsterdam, 1967) 313–321.
20 M.W. McElhinny and N.D. Opdyke, The palaeomagnetism of the Precambrian dolerites of eastern Southern Rhodesia, an example of geologic correlation by rock magnetism, J. Geophys. Res. 69 (1964) 2465–2475.
21 G.S. Watson, A test for randomness of directions, Mon. Not. R. Astron. Soc., Geophys. Suppl. 7 (1956) 160–161.

22 S.A. Vincenz and J.McG. Bruckshaw, Note on the probability distribution of a small number of vectors, Proc. Camb. Phil. Soc. 56 (1960) 21–26.
23 M.W. McElhinny and P.J. Burek, Mesozoic palaeomagnetic stratigraphy, Nature 232 (1971) 98–102.
24 M.W. McElhinny, J.C. Briden, D.L. Jones and A.B. Brock, Geological and geophysical implications of palaeomagnetic results from Africa, Rev. Geophys. 6 (1968) 201–238.
25 M.W. McElhinny and A. Brock, A new palaeomagnetic result from East Africa and estimates of the Mesozoic palaeoradius, Earth Planet. Sci. Lett. 27 (1975) 321–328.
26 D.H. Tarling, Another Gondwanaland, Nature 238 (1972) 92–93.
27 P.E. Kent, East African evidence of the palaeoposition of Madagascar, in: Implications of Continental Drift to the Earth Sciences, eds. D.H. Tarling and S.K. Runcorn (Academic Press, 1973) 873–875.
28 L.C. King, Basic palaeo-geography of Gondwanaland during the Late Palaeozoic and Mesozoic eras, Q. J. Geol. Soc. London CXIV (1958) 47–70.
29 F. Dixey, The geology and geomorphology of Madagascar, and a comparison with eastern Africa, Q. J. Geol. Soc. London CXVI (1959) 255–268.
30 J.H. Wellington, Southern Africa – A Geographical Study (Cambridge Univ. Press, 1955) 528 pp.
31 R.V. Dingle and H.C. Klinger, Significance of Upper Jurassic sediments in the Knysna outlier (Cape Province) for timing of the breakup of Gondwanaland, Nature Phys. Sci. 232 (1971) 37–38.
32 S.H. Haughton, Stratigraphic History of Africa South of the Sahara, (Oliver and Boyd, London, 1963) 365 pp.
33 M.W. McElhinny, Palaeomagnetism and Plate Tectonics, (Cambridge Univ. Press, 1973) 358 pp.
34 J.R. Heirtzler and R.H. Burroughs, Madagascar's palaeoposition: New data from the Mozambique Channel, Science 174 (1971) 488–490.
35 A.S. Laughton, D.H. Matthews and R.L. Fisher, The structure of the Indian Ocean, in: The Sea, ed. A.E. Maxwell (Wiley, 1970) vol. 4, part II: 543–586.
36 L.O. Nicolaysen, South African studies of the geodynamics of continental and oceanic rifts, in: International Geodynamics Project, Plan for South African Participation (C.S.I.R., Pretoria, 1973).
37 R.A. Scrutton, Structure and evolution of the sea floor south of South Africa, Earth Planet. Sci. Lett. 19 (1973) 250–256.
38 K. Burke and A.J. Whiteman, Uplift, rifting and the breakup of Africa, in: Implications of Continental Drift to the Earth Sciences, eds. D.H. Tarling and S.K. Runcorn (Academic Press, 1973) 735–755.
39 J.B. Wright and P. McCurry, A reply to "The significance of sandstone inclusions in lavas of the Comores Archipalago" by M.F.J. Flower and D.F. Strong, Earth Planet. Sci. Lett. 8 (1970) 267–268.
40 M.W. McElhinny, Formation of the Indian Ocean, Nature 228 (1970) 977–979.
41 B.P. Luyendyk, Gondwanaland dispersal and the early formation of the Indian Ocean, in: T.A. Davies, B.P. Luyendyk et al., Initial Reports of the Deep Sea Drilling Project 26 (1974) 945–952.